

CRR Binomial Option Pricing Pipeline

Full Build Pipeline Flowchart

A six-page walkthrough of every component across the three pipeline layers:

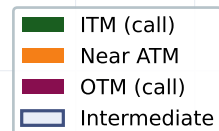
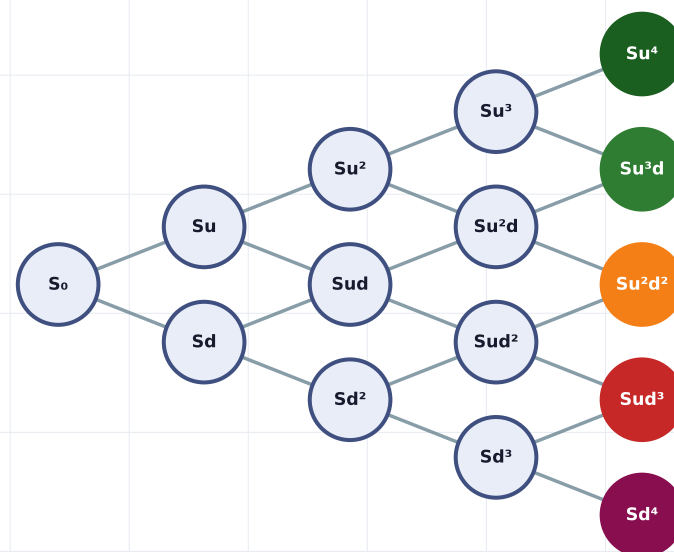
- (L1) CRR Pricing Engine
- (L2) Prediction Market ML Extensions
- (L3) Cross-Model Comparison

L1 CRR Pricing Engine

L2 ML Extensions

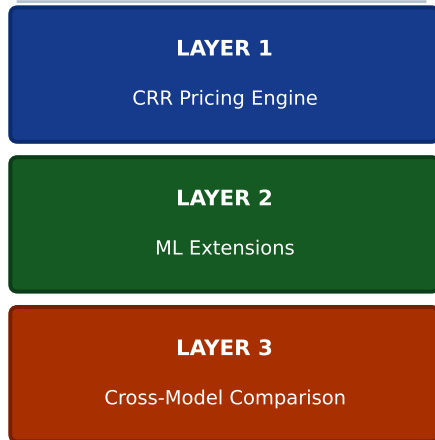
L3 Cross-Model Comparison

CRR Binomial Tree ($N = 4$)



Build Pipeline

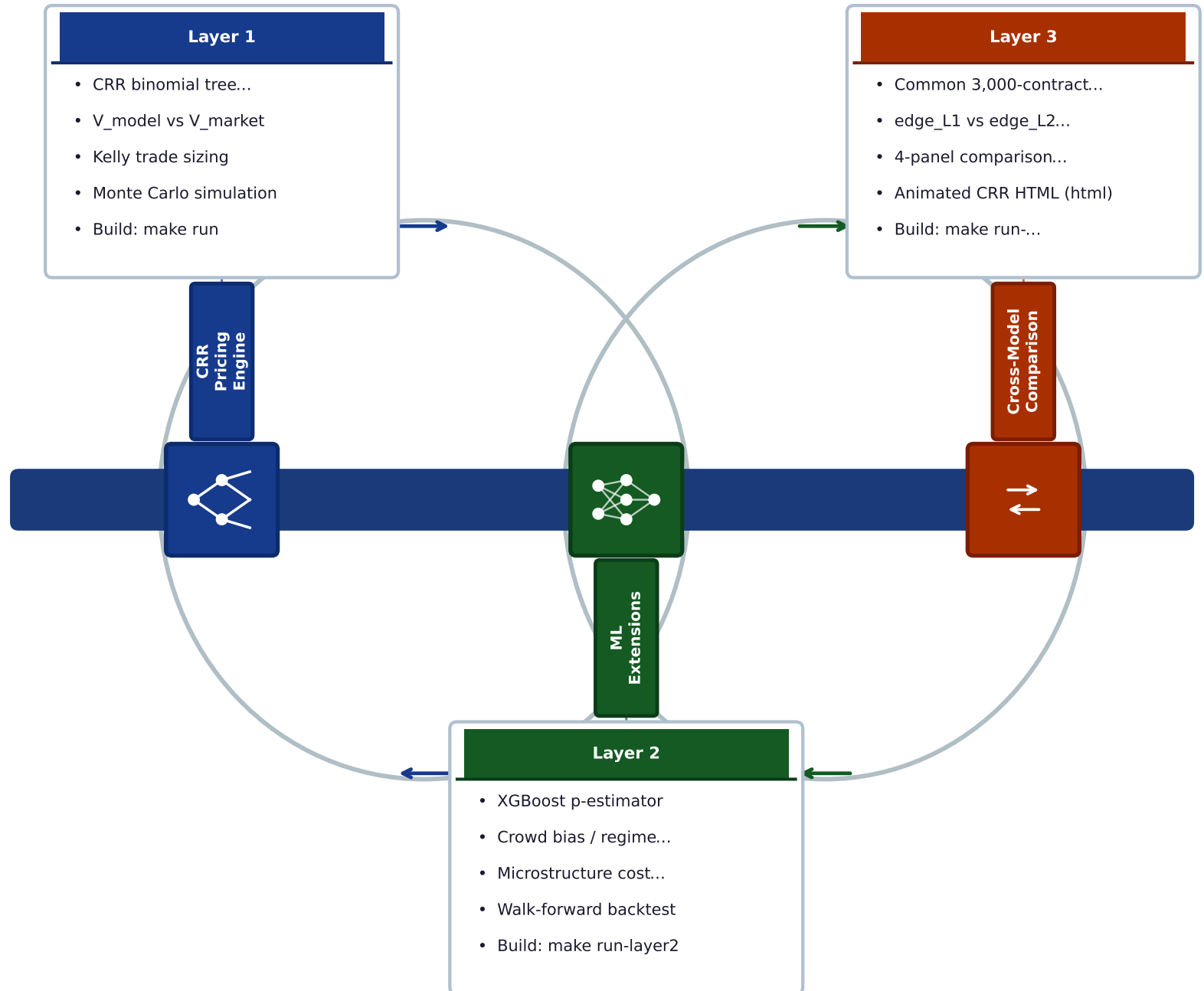
Three independent layers connected by shared edge signals.



make run-all

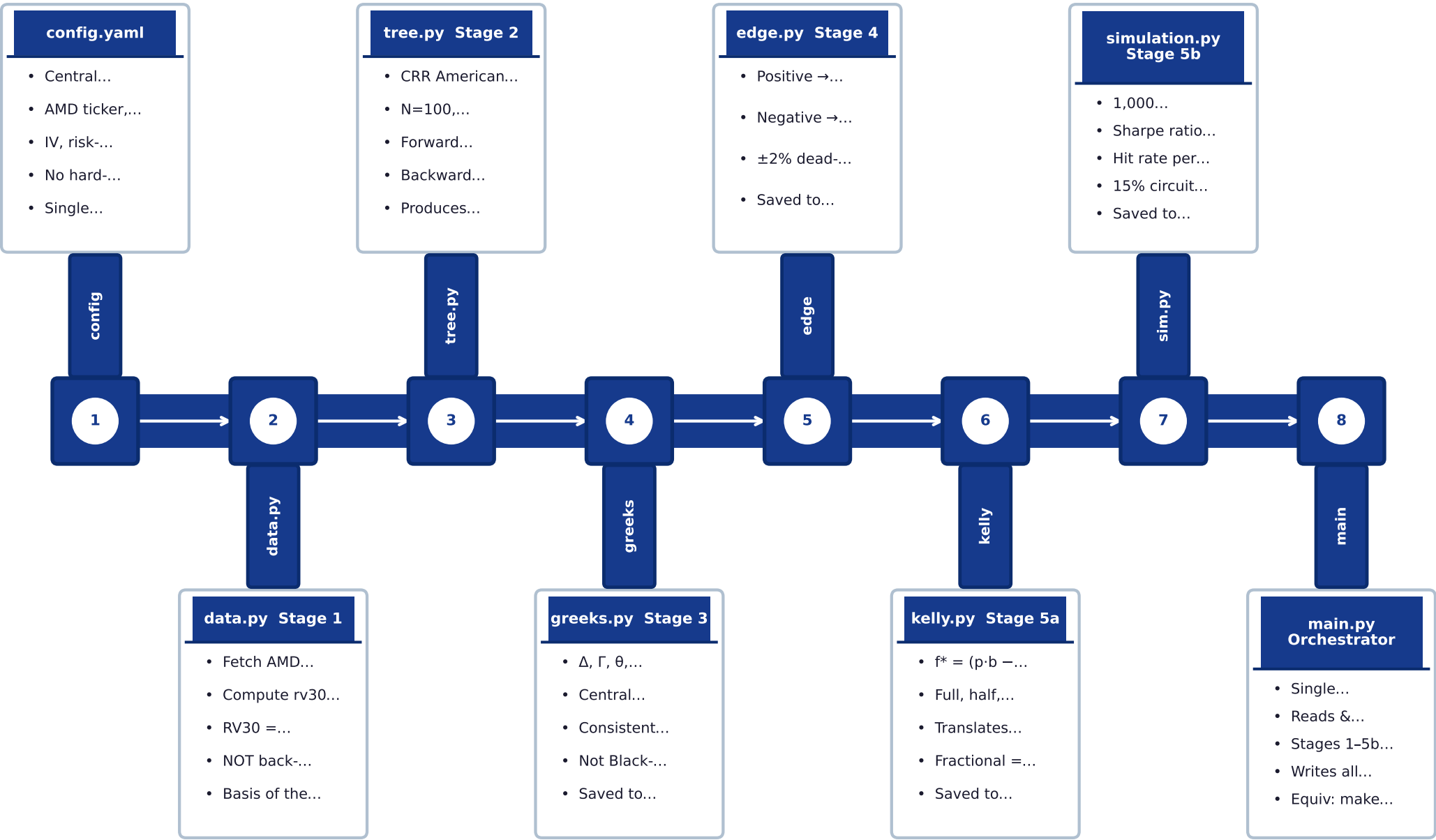
Chains all three layers in sequence automatically.

L1 and L2 run independently; L3 merges both edge signals on the same AAPL sample



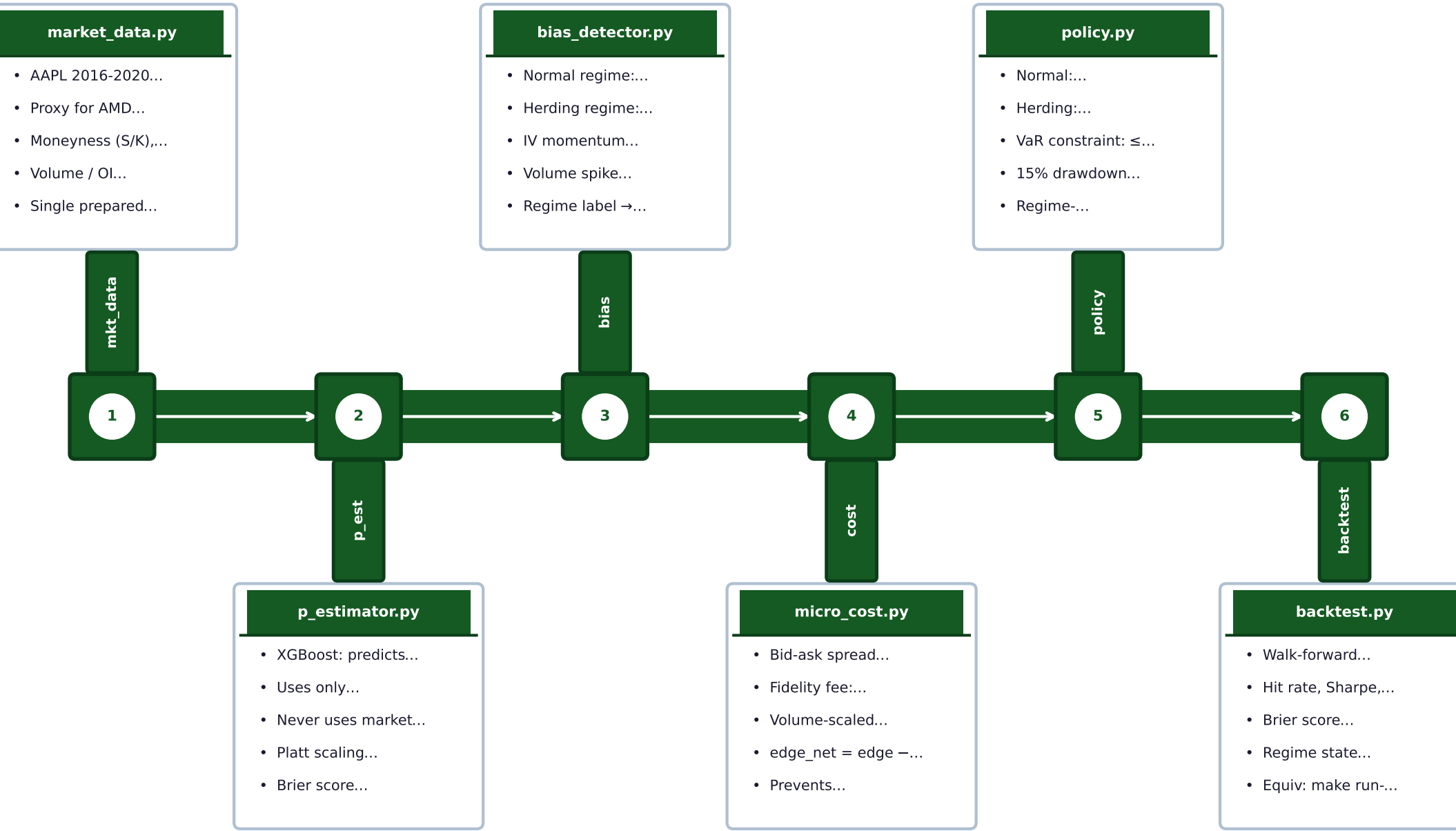
Layer 1 — CRR Pricing Engine: Component Flow

Build: make run · Entry: python main.py config.yaml · Outputs: project/outputs/



Layer 2 — Prediction Market Extensions: Component Flow

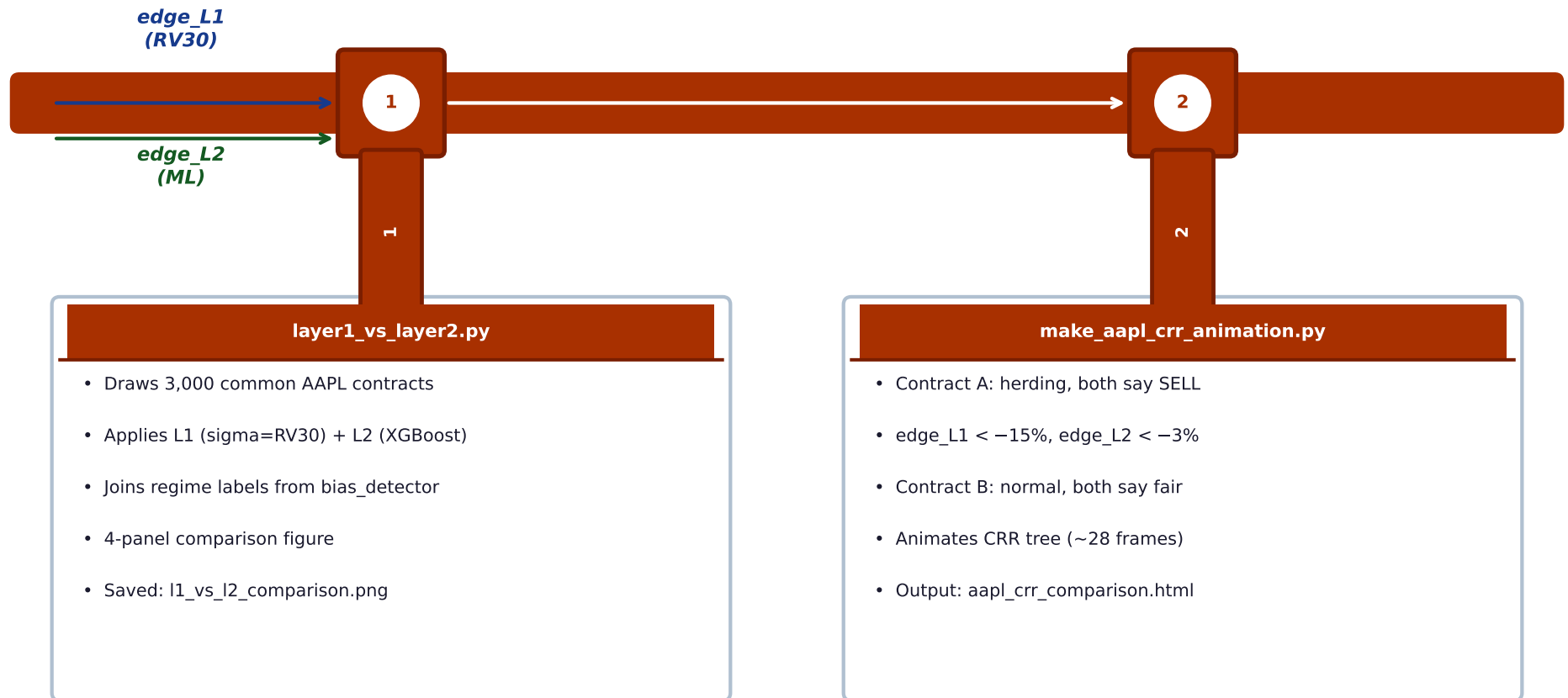
Build: make run-layer2 · Data: AAPL 2016-2020 Kaggle · Outputs: project/outputs/



make run-comparison · Outputs: l1_vs_l2_comparison.png + aapl_crr_comparison.html

Core Scientific Question

Do CRR (mathematical, no-arbitrage) and XGBoost (empirical, ML) independently agree on which AAPL options the crowd misprices? Agreement validates both models simultaneously; disagreement zones are findings in their own right.



Run from project root · Sequence: setup → run → run-layer2 → run-comparison

make setup	make run	make run-layer2	make run-comparison
• Create .venv • Install Poetry • Install dependencies • Run once	• main.py config.yaml • All 6 Layer 1 stages • Outputs →... • Layer 1 complete	• Runs 6 Layer 2 modules • Trains XGBoost models • Outputs →... • Layer 2 complete	• Runs L3 comparison... • Requires L1 + L2 first • Outputs: PNG + HTML • Layer 3 complete

